

#1 Considere la suma $S_n = \sum_{k=1}^n \frac{k}{2^k}$.

a) Demuestre por inducción que $S_n = 2 - \frac{n+2}{2^n}$, para todo $n \geq 1$.

$$n=1 \quad \frac{1}{2^1} = 2 - \frac{3+2}{2^1}$$
$$\frac{1}{2} = \frac{1}{2} \quad \checkmark$$

$$n=p \quad \sum_{k=1}^p \frac{k}{2^k} = 2 - \frac{p+2}{2^p}, \text{ H.I.}$$

$$n=p+1 \quad \sum_{k=1}^{p+1} \frac{k}{2^k} = 2 - \frac{(p+1)+2}{2^{p+1}}$$

$$2 - \frac{p+3}{2^{p+1}}$$

$$2 + \frac{-p-3}{2^{p+1}}$$

Demonstracion

$$\begin{aligned} \textcircled{P+1} \quad \sum_{k=1}^{\infty} \frac{k}{2^k} & \quad \begin{matrix} P+1 & P \\ \sum_{k=0}^{\infty} n = \sum_{k=0}^{\infty} n + (P+1) \end{matrix} \end{aligned}$$

$$\sum_{k=1}^{\infty} \frac{k}{2^k} + \frac{P+1}{2^{P+1}}$$

$$2 - \frac{P+2}{2^P} + \frac{P+1}{2^{P+1}}, \text{ H.I.} \quad \sum_{k=1}^P \frac{k}{2^k} = 2 - \frac{P+2}{2^P}, \text{ H.I.}$$

$$2 + \frac{P+1}{2^{P+1}} - \frac{P+2}{2^P} - \frac{2^1}{2^2}$$

$$2 + \frac{P+1}{2^{P+1}} - \frac{(P+2)2}{2^{P+1}}$$

$$2 + \frac{P+1 - [2P+4]}{2^{P+1}}$$

$$2 + \frac{P+1-2P-4}{2^{P+1}}$$

$$2 + \frac{-P-3}{2^{P+1}}$$

$$\boxed{2 - \frac{P+3}{2^{P+1}}}$$

$$\begin{array}{ccc}
 p+1 & & p \\
 \leq & 2^h = & \leq 2^h + 2^{h+1} \\
 h=1 & & h=1
 \end{array}$$

b) Determine si la serie $\sum_{k=1}^{\infty} \frac{k}{2^k}$ converge y, en caso afirmativo, halle su suma. $\rightarrow S_n = 2 - \frac{n+2}{2^n}$

$$\lim_{n \rightarrow \infty} 2 - \frac{n+2}{2^n} = 2 \quad \text{Converge,} \quad \text{Suma} = 2$$

$$n^p < a^n$$

#2 Considere la serie $\sum_{k=1}^{\infty} \frac{1}{r^{k+1}}$, donde r es una constante real, $r \neq 0$. Sabiendo que la serie es convergente y su suma es $S = \frac{1}{2}$, determine los posibles valores de r . (3 Pts) $n=k$

$$\begin{array}{c}
 \infty \\
 \sum_{h=1}^{\infty} \frac{1}{r^{h+1}}
 \end{array}$$

$$\begin{array}{c}
 \infty \\
 \sum_{h=1}^{\infty} \frac{1}{r^h \cdot r} \rightarrow r^{h+1}
 \end{array}$$

$$\begin{array}{c}
 \infty \\
 \frac{1}{r} \leq \frac{1}{r^h}
 \end{array}$$

$$\begin{array}{c}
 \infty \\
 \frac{1}{r} \leq \frac{1^n}{r^n}
 \end{array}$$

$$\frac{1}{r} \sum_{h=1}^{\infty} \frac{r^h}{r^h}$$

$$\frac{1}{r} \sum_{h=1}^{\infty} \left(\frac{1}{r} \right)^h \quad \frac{r^p}{1-r} = \frac{1}{2}$$

↳ r

$$\frac{1}{r} \frac{\left(\frac{1}{r} \right)^2}{1 - \frac{1}{r}} = \frac{1}{2}$$

$$\frac{1}{r} \frac{\frac{1}{r}}{\frac{r-1}{r}} = \frac{1}{2}$$

$$\frac{1}{\cancel{r}} \cdot \frac{\cancel{r}}{r^2 - r} = \frac{1}{2}$$

$$\frac{1}{r^2 - r} = \frac{1}{2}$$

$$2 = r^2 - r$$

$$r^2 - r - 2 = 0$$

$$\begin{array}{rcl} r & \times & 1 = r \\ r & \times & 2 = -2r \\ & & -r \end{array}$$

$$(r+1)(r-2) = 0$$

$$\boxed{r = -1 \quad r = 2}$$

#3 Para cada una de las siguientes series, analice y justifique si converge o diverge.

$$a) \sum_{k=2}^{\infty} \frac{\arctan(k) + \pi}{\sqrt[3]{k^2 - 1}} \quad n=15 \quad \begin{aligned} \sin, \cos &= -1 \leq x \leq 1 \\ \sin^2, \cos^2 &= 0 \leq x \leq 1 \\ \arctan &= -\frac{\pi}{2} < x < \frac{\pi}{2} \end{aligned}$$

$$-\frac{\pi}{2} < \arctan(k) < \frac{\pi}{2}$$

$$-\frac{\pi}{2} + \pi < \arctan(k) + \pi < \frac{\pi}{2} + \pi$$

$$\frac{-\frac{\pi}{2} + \pi}{\sqrt[3]{k^2 - 1}} < \frac{\arctan(k) + \pi}{\sqrt[3]{k^2 - 1}} < \frac{\frac{\pi}{2} + \pi}{\sqrt[3]{k^2 - 1}}$$

∞

∞

$$\sum_{k=2}^{\infty} \frac{\frac{\pi}{2} + \pi}{\sqrt[3]{k^2 - 1}} \approx \sum_{k=2}^{\infty} \frac{1}{\sqrt[3]{k^2}} = \frac{1}{k^{\frac{2}{3}}}, \quad \begin{aligned} &\text{p serie} \\ &p \leq 1 \end{aligned}$$

$\therefore$ Divergente

$$a) \sum_{k=2}^{\infty} \frac{\arctan(k) + \pi}{\sqrt[3]{k^2 - 1}} \quad \therefore \text{Es divergente}$$

#3 Para cada una de las siguientes series, analice y justifique si converge o diverge.

$$b) \sum_{k=1}^{\infty} \frac{k^2 + 7^{-k}}{3^k \cdot (k^2 + k)} \quad n=15$$

∞

∞

$$\sum_{k=1}^{\infty} \frac{k^2}{3^k \cdot (k^2 + k)} + \sum_{k=1}^{\infty} \frac{7^{-k}}{3^k \cdot (k^2 + k)}$$

∞

∞

$\frac{1}{7^k}$

$$\sum_{k=1}^{\infty} \frac{k^2}{3^k \cdot k^2} + \sum_{k=1}^{\infty} \frac{1}{3^k \cdot k^2}$$

$$\sum_{k=1}^{\infty} \frac{1}{3^k} \checkmark + \sum_{k=1}^{\infty} \frac{1}{2 \cdot 3^k \cdot k^2}$$

$$c) \sum_{k=1}^{\infty} \frac{\ln(k) + 2^k}{3^k + k^2} \quad \begin{array}{l} \swarrow a_n \\ \text{Por criterio de comparacion} \\ \text{a) limite} \end{array}$$

$$\ln(n) < n^p < a^n$$

$$\ln(n) < n^2 < 2^n$$

∞

$$\sum_{n=1}^{\infty} \frac{2^n}{3^n}$$

∞

$$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n \quad \left|\frac{2}{3}\right| = \frac{2}{3} < 1$$

Converge

$\nwarrow b_n$

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$L = \pi, \pi \neq 0, a_n$ converge o diverge
 Si b_n converge o diverge

$L = 0 \wedge b_n$ converge $\rightarrow a_n$ converge

$L = \infty \wedge b_n$ diverge $\rightarrow a_n$ diverge

$$\lim_{n \rightarrow \infty} \frac{\cancel{\ln(n)} + 2^n}{3^n + \cancel{n^2}} \quad \text{Por dominancia}$$

$$\frac{2^n}{3^n}$$

$$\lim_{n \rightarrow \infty} \frac{2^n}{3^n} = \frac{2}{3}$$

$L \neq 0 \wedge b_n$ converge
 si a_n converge

$$d) \sum_{n=1}^{\infty} \left(\frac{-n+5}{2n+9} \right)^{2n+1}$$

$(r)^n \rightarrow \text{Geometrica}$

$(\text{Expression})^n \rightarrow \text{Rais}$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{-n+5}{2n+9} \right)^{2n+1}}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{p}, p \text{ constante} = 1$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left[\left(\frac{-n+5}{2n+9} \right)^{2n} \cdot \left(\frac{-n+5}{2n+9} \right)^1 \right]}$$

$$\sqrt[n]{p(x)} = 1$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left[\left(\frac{-n+5}{2n+9} \right)^2 \right]^n} \cdot \sqrt[n]{\frac{-n+5}{2n+9}}$$

$$\sqrt[n]{\frac{p(x)}{q(x)}} = 1$$

$$\lim_{n \rightarrow \infty} \left(\frac{-n+5}{2n+9} \right)^2 \quad K < n^p$$

$$\sqrt[n]{K} = 1$$

$$\lim_{n \rightarrow \infty} \left(\frac{-n}{2n} \right)^2$$

$$\sqrt[n]{n^n} = n$$

$$\left(\frac{1}{2} \right)^2 = \frac{1}{4} = 0,25 < 1$$

$\therefore$ Convergente

$$(n+2)! = (n+2)(n+1)n(n-1)!\dots$$

7. Considere la serie de potencias

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n!} x^{3n} \rightarrow \text{Razon}$$

Determine el intervalo de convergencia de esta serie (no estudie los extremos del intervalo).

[3 pts]

$$\lim_{n \rightarrow +\infty} \frac{(-1)^{n+1} \cdot x^{3n+3}}{(n+1)!} \cdot \frac{n!}{(-1)^n \cdot x^{3n}}$$

$$\lim_{n \rightarrow +\infty} \frac{(-1)^{n+1} \cdot |x|^{3n+3}}{(n+1)!} \cdot \frac{n!}{(-1)^n \cdot x^{3n}}$$

$$|x|^3 \lim_{n \rightarrow +\infty} \frac{1}{n+1} \quad L = \frac{a_{n+1}}{a_n}$$

$$I = \mathbb{R}$$

$$\begin{aligned} L &= \frac{1}{2} \quad L \cdot |x-c| < 1 \\ |x-c| &< \frac{1}{L} \\ -\frac{1}{L} &< x-c < \frac{1}{L} \\ -\frac{1}{L} + c &< x < \frac{1}{L} + c \\ I &= \left] -\frac{1}{L} + c, \frac{1}{L} + c \right[\end{aligned}$$

$$L=0 \quad L \cdot |x-c| < 1 \quad I = \mathbb{R}$$

$$\begin{aligned} L &= +\infty \quad +\infty |x-c| < 1 & x &= 2 \\ & & x &= 2 \\ & & I &: x=2 \end{aligned}$$

$$I: x=c$$

4. Utilice el criterio de la integral para determinar si la serie $\sum_{k=6}^{\infty} \frac{1}{k \ln^2 k}$ converge o diverge.

[4 pts]

1) $f(x)$ 2) $\int_6^{\infty} f(x)$ $\rightarrow$ converge
 $\rightarrow$ too diverge

1) $f(x) = \frac{1}{x \ln^2(x)}$ $\left[\frac{1}{f(x)} \right]' = \frac{-1}{f^2(x)} f'(x)$

$$= \frac{-1}{(x \ln^2(x))^2} \cdot \ln^2(x) + x \cdot \{ \ln(x) \}^{2 \cdot 1}$$

$$= \frac{-1}{(x \ln^2(x))^2} \cdot \ln^2(x) + \cancel{x} \cdot 2 \cdot \ln(x) \cdot \frac{1}{\cancel{x}}$$

$$= \frac{-1}{(x \ln^2(x))^2} \cdot \ln^2(x) + \ln(x)$$

$\therefore$ Decrease

$$\sum_{n=6}^{\infty} \frac{1}{n \ln^2(n)} \rightarrow \int_6^{\infty} \frac{1}{x \ln^2(x)} dx$$

$$\rightarrow \int \frac{1}{x \ln^2(x)} dx \quad u = \ln(x) \\ du = \frac{1}{x} dx$$

$$\int \frac{1}{u^2} du$$

$$\int \frac{1}{u^2} du$$

$$\int u^{-2} = \frac{u^{-1}}{-1} = -\frac{1}{u} = -\frac{1}{\ln(x)}$$

$$\frac{-1}{\ln(x)} \Big|_6^{+\infty}$$

$$\lim_{x \rightarrow +\infty} \frac{-1}{\ln(x)} = -\frac{1}{\ln(6)}$$

$$\frac{1}{\ln(6)}, \text{ } \therefore \text{ converge}$$

$$\sum_{k=6}^{\infty} \frac{1}{k \ln^2 k} \quad \therefore \text{ converge por el criterio de la integral}$$

Series Alternadas

Para que converja

1) a_n sea decreciente

$$\left\{ \begin{array}{l} f'(x) \\ \frac{a_{n+1}}{a_n} \geq 1 \end{array} \right. \begin{array}{l} \checkmark \rightarrow \text{crece} \\ \times \rightarrow \text{decreciente} \end{array}$$

2) $\lim_{n \rightarrow \infty} a_n = 0$

Convergencia Absoluta y Condicional

Error de Series Alternadas

$\sum_{n=1}^{\infty} (-1)^n \frac{2^n}{n!}$, que el error sea menor que 10^{-3}

N	$N+1$	$a_n < 0,001$
$\downarrow$	$\downarrow$	$\downarrow$
$ R $	Prueba	Serie

6	7	$\frac{2^7}{7!} \approx 0,025 \not< 0,001$
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9	10	$\frac{2^{10}}{10!} \approx 0,0002 < 0,001$
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$N = 9$

1. Considere la sucesión con término énsimo dado por $x_n = \frac{n!}{2023^n}$. Determine a partir de cuál término dicha sucesión se hace creciente.

$$\frac{(n+1)!}{2023^{n+1}} \geq \frac{n!}{2023^n}$$

$$\frac{a_{n+1}}{a_n} \geq 1$$

$$\frac{(n+1) \cdot \cancel{n!} \cdot \cancel{2023^n}}{\cancel{n!} \cdot \cancel{2023^n} \cdot 2023^1} \geq 1$$

$$\frac{n+1}{2023} \geq 1$$

$$n+1 \geq 2023$$

$$n \geq 2022$$

2. Analice y justifique la convergencia de la siguiente serie, y calcule su suma

$$\sum_{k=3}^{\infty} \frac{9 \cdot 5^{-k}}{(-3)^{2-k}} - \frac{2}{k^2 - 1}$$

∞

∞

$$\sum_{k=3}^{\infty} \frac{9 \cdot 5^{-k}}{(-3)^{2-k} \cdot (-3)^{-k}} - \sum_{k=3}^{\infty} \frac{2}{k^2 - 1}$$

$$= \left(\frac{-3}{5} \right)^k - \frac{2}{(k-1)(k+1)} = \frac{A}{k-1} + \frac{B}{k+1}$$

$$|r| = \left| \frac{-3}{5} \right| = \frac{3}{5} < 1 \quad \text{---} \quad 2 = A(k+1) + B(k-1)$$

$$\therefore \text{Converge} \quad \text{---} \quad k=1 \rightarrow 2 = 2A \rightarrow A=1$$

$$k=-1 \rightarrow 2 = -2B \rightarrow B=-1$$

$$\frac{\left(\frac{-3}{5} \right)^3}{1 - \frac{3}{5}} = \frac{-27}{500}$$

$$\frac{2}{(k-1)(k+1)} = \frac{A}{k-1} + \frac{B}{k+1} \quad \begin{matrix} A=1 \\ B=-1 \end{matrix}$$

$$\frac{-27}{200} - \frac{\infty}{\infty} \quad \frac{1}{k-1} - \frac{1}{k+1}$$

$k=3$

$$\frac{\infty}{\infty} \quad \frac{1}{k-1} - \frac{1}{k} + \frac{1}{k} - \frac{1}{k+1}$$

$k=3$ $k=3$

$$\frac{1}{2} - \lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{3} - \lim_{n \rightarrow \infty} \frac{1}{n+1}$$

$$\frac{-27}{200} + \frac{5}{6}$$

$$\frac{719}{600}$$

$$\sin, \cos \quad -1 \leq \sin, \cos \leq 1$$

$$\sin^2, \cos^2 \quad 0 \leq \sin^2, \cos^2 \leq 1$$

$$\arctan \quad -\frac{\pi}{2} < \arctan < \frac{\pi}{2}$$

$$\arccos \quad 0 \leq \arccos \leq \pi$$

! 0 ... $\rightarrow$ Crit de razon

3. Para cada una de las siguientes series, analice y justifique si es convergente o divergente

a) $\sum_{n=2}^{\infty} \frac{\ln(n)}{n^2}$

$$f(x) = \frac{\ln(x)}{x^2}$$

$$f'(x) = \frac{\frac{1}{x} \cdot x^2 - \ln(x) \cdot 2x}{x^4}$$

$$= \frac{x - 2x \ln(x)}{x^4}$$

$$= \frac{x(1 - 2\ln(x))}{x^4} = 0$$

$$\frac{1 - 2\ln(x)}{x^3} = 0 \quad f \nexists x=0$$

$$1 - 2\ln(x) = 0$$

$$1 = 2\ln(x)$$

$$\ln(x) = \frac{1}{2}$$

$$x = e^{\frac{1}{2}}$$

$-\infty$		0	$e^{\frac{1}{2}}$	$+\infty$
x^3	-	•	+	+
$1 - 2\ln(x)$	+	+	•	-
$f'(x)$	-	+	-	-
$f(x)$	↗	↗	↘	

Decrease

$$2 \int_{-\infty}^{+\infty} \frac{\ln(x)}{x^2} \rightarrow \int \ln(x) \cdot x^{-2} \quad \begin{array}{l} u = \ln(x) \quad dv = x^{-2} \\ du = \frac{1}{x} dx \quad v = \frac{-1}{x} \end{array}$$

$$\rightarrow \frac{-\ln(x)}{x} - \int \frac{-1}{x} \cdot \frac{1}{x} dx$$

$$\frac{-\ln(x)}{x} + \int x^{-2}$$

$$\frac{-\ln(x)}{x} - \frac{1}{x}$$

$$\frac{-\ln(x) - 1}{x}$$

$$\begin{array}{l} \ln(x) < x^p \\ \ln(x) < x^p \end{array}$$

$$\lim_{x \rightarrow +\infty} \frac{\ln(x) - 1}{x} = \frac{\ln(2) + 1}{2}$$

$$\frac{\ln(2) + 1}{2}$$

is Converge

$$\begin{array}{l} \frac{f(x)}{g(x)} \quad f(x) > g(x) = +\infty \quad \frac{n!}{n} \\ f(x) < g(x) = 0 \quad \frac{n}{n!} \end{array}$$